

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT2 — VISUALIZATION ANALYSIS REPORT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Evaluation (BL5)	Assessment:	Assessment Tool 2 – Visualization Analysis Report
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	<i>Evaluate the effectiveness of visualization design choices — proportional ink, axis scales, overplotting solutions, color use, and accessibility — and justify improvements.(BL5)</i>		
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Section / Batch:	2023/AI&DS	Date:	30.8.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond describing the scenario: explain WHY the design choice is flawed, and justify your judgment.
- Where a question asks for a fix, propose a specific improvement — not just “make it better.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION ANALYSIS REPORT

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT2 — Visualization Analysis Report

CO5 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO5-AT2 Visualization Analysis Report problems, in which students critique a described visualization design choice, identify its specific flaw, and justify a well-reasoned improvement.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant visualization principle being evaluated (e.g., proportional ink, axis scale choice, overplotting, color scale design, accessibility).	Good understanding of the relevant principle, with only minor gaps.	Basic understanding, but with some misconceptions about which principle applies.	Poor understanding of core principles; misidentifies the nature of the flaw entirely.
Explanation	25%	Clearly explains why the design choice is flawed or effective, including its practical consequence for a reader — not just that it is 'good' or 'bad.'	Explains the evaluation with adequate reasoning, covering most of the practical consequence.	Identifies the issue but explanation is generic or only loosely tied to its practical effect.	Little or no explanation provided; the evaluation is a bare assertion.
Application	20%	Proposes a specific, well-justified fix or recommendation that directly addresses the evaluated issue.	Proposes a workable fix, with only minor gaps in justification.	Proposes a fix, but it only partially addresses the issue or lacks justification.	Fails to propose a workable fix, or it does not address the evaluated issue.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Organization	15%	The response is clearly structured, addressing critique and recommendation in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	10%	Both the critique and the recommendation are completely accurate and well-suited to the specific scenario described.	Mostly accurate, with only minor omissions or a small error.	Partially correct; the critique or recommendation contains notable inaccuracies.	Largely incorrect or inappropriate, with major gaps in both the critique and the recommendation.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

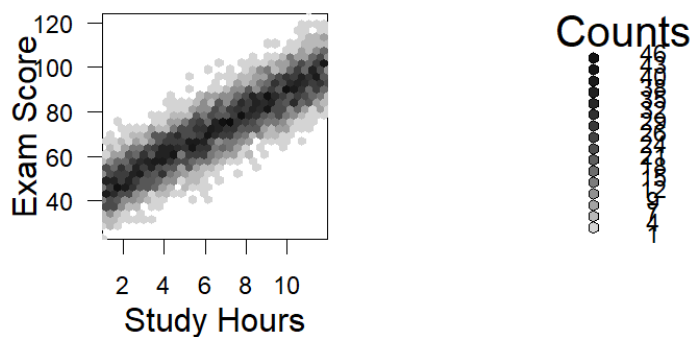
Problem 21

Data: A dataset of 6,500 (study hours, exam score) point pairs is too dense to read as a raw scatterplot.

Task: Explain what each cell's color would represent in a 2D histogram of this data, and what a very dark cell would indicate.

```
1 # Generate sample data
2 set.seed(123)
3 study_hours <- runif(6500, 1, 12)
4 exam_score <- 40 + 5 * study_hours + rnorm(6500, 0, 8)
5 # Create 2D histogram
6 hist2d <- hist(
7   study_hours,
8   breaks = 30,
9   plot = FALSE
10 )
11 # Use hexbin for a clearer 2D binned visualization
12 library(hexbin)
13 hex <- hexbin(
14   study_hours,
15   exam_score,
16   xbins = 30
17 )
```

2D Histogram: Study Hours vs Exam Score



In a **2D histogram**, the x-axis represents **study hours** and the y-axis represents **exam score**. The plot divides the data space into cells or bins.

- Each cell's **color intensity represents the number of observations** falling inside that cell.
- A **light-colored cell** contains relatively few students.
- A **very dark cell** contains a large number of observations.
- Therefore, a dark cell identifies a region where many students have a similar combination of study hours and exam scores.

A 2D histogram is useful here because 6,500 points would cause severe **overplotting** in a normal scatterplot.

Problem 22

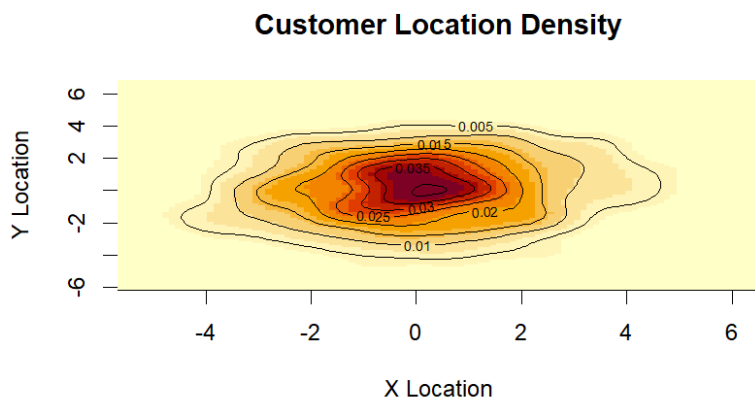
Data: A density plot of a dense cluster of customer locations around a city center is shown with contour lines overlaid.

Task: Explain what tightly-packed contour lines would indicate compared to widely-spaced lines in this context.

- **Tightly packed contour lines** indicate that density is changing rapidly over a short distance. This represents a concentrated or sharply defined customer cluster.
- **Widely spaced contour lines** indicate a more gradual change in density and therefore a more spread-out customer distribution.
- The center of the tightly concentrated contours generally represents a region with a **higher concentration of customers**.

Thus, contour spacing helps us understand how quickly customer density changes across the city.

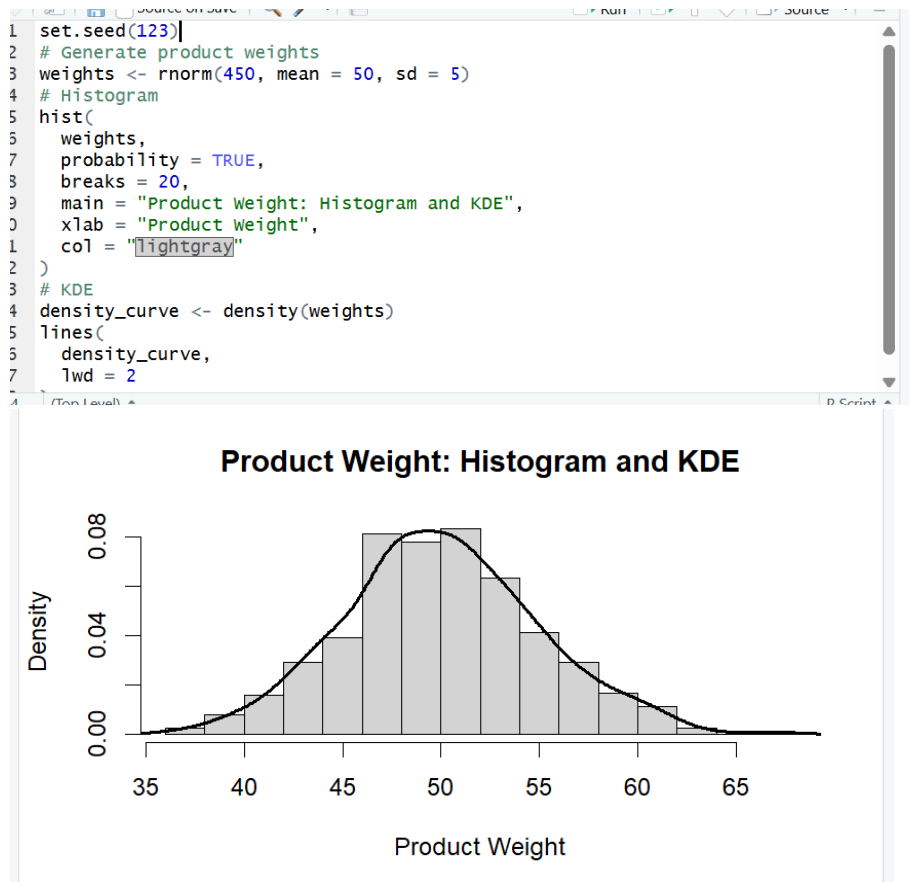
```
1 set.seed(123)
2 # Generate customer locations
3 x <- rnorm(1000, mean = 0, sd = 2)
4 y <- rnorm(1000, mean = 0, sd = 2)
5 # Create 2D density estimate
6 library(MASS)
7 density_data <- kde2d(x, y, n = 100)
8 # Plot density
9 image(
10   density_data,
11   main = "Customer Location Density",
12   xlab = "X Location",
13   ylab = "Y Location"
14 )
15 # Add contour lines
16 contour(
17   density_data,
```



Problem 23

Data: A dataset of 450 product weight measurements is shown as both a histogram and a KDE.

Task: Explain one advantage the KDE has over the histogram, and describe what would happen if the KDE's bandwidth were set too wide.



One advantage of a KDE (Kernel Density Estimate) over a histogram is that KDE

produces a smooth estimate of the underlying distribution.

A histogram depends strongly on the selected bin width and bin boundaries, while KDE provides a continuous curve that makes the overall shape easier to observe.

If the KDE bandwidth is set too wide:

- The density curve becomes excessively smooth.
- Important peaks and small variations disappear.
- Different groups in the data may be merged together.
- The resulting plot can hide meaningful structure in the product-weight distribution.

Therefore, bandwidth should be chosen carefully.

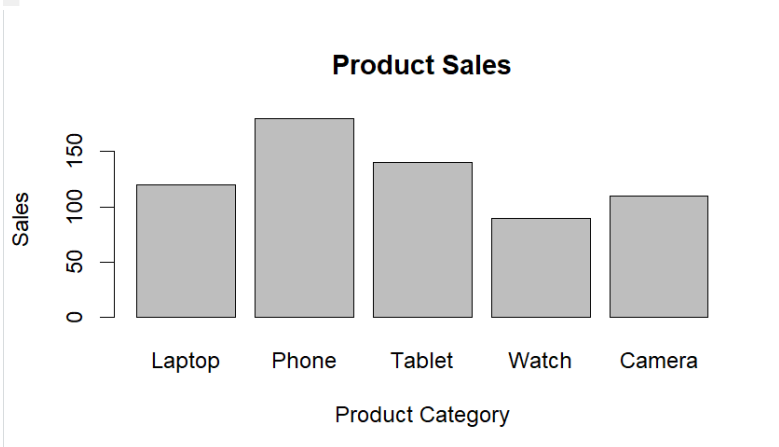
Problem 24

Data: A bar chart shows 5 product categories, colored with clashing neon hues.

Task: Critique this color choice and propose a more purposeful, consistent alternative. The neon colors are a poor design choice because they are visually distracting and do not communicate meaningful information.

```
products <- c("Laptop", "Phone", "Tablet", "Watch", "Camera")
sales <- c(120, 180, 140, 90, 110)
```

```
# Original chart with different colors
barplot(
  sales,
  names.arg = products,
  main = "Product Sales",
  xlab = "Product Category",
  ylab = "Sales"
)
```



Using five unrelated bright colors can:

- Distract attention from the actual sales values.

- Make the chart look cluttered.
- Reduce visual consistency.
- Create accessibility problems for some viewers.
- Suggest that the colors represent different meanings when they do not.

A better solution is to use a single consistent color when all categories represent the same variable.

If color is required to highlight one important category, a neutral palette can be used with one contrasting color for emphasis.

Therefore, color should communicate information rather than simply decorate the chart.

Problem 25

Data: A chart displays 10 data points, encoding 6 variables including an irrelevant internal ID number.

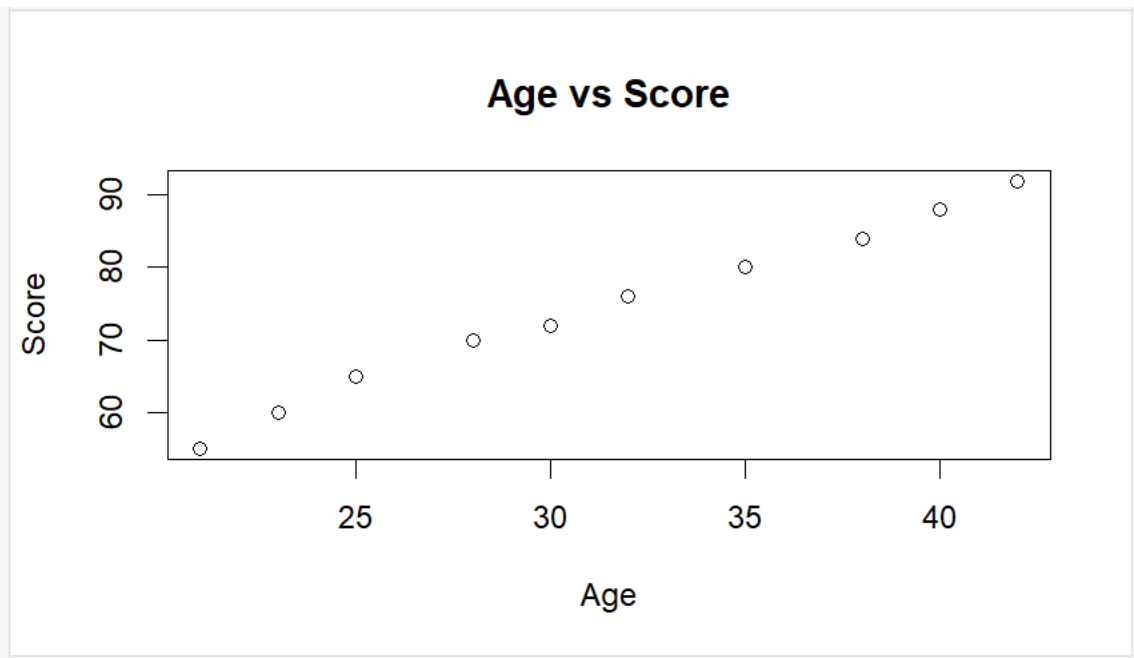
Task: Evaluate whether this design is effective for a general audience, and propose a simplification.

```

1 # Example dataset
2 data <- data.frame(
3   ID = 1001:1010,
4   Age = c(21, 23, 25, 28, 30, 32, 35, 38, 40, 42),
5   Income = c(25, 30, 35, 42, 48, 52, 60, 68, 75, 80),
6   Score = c(55, 60, 65, 70, 72, 76, 80, 84, 88, 92),
7   Experience = c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10),
8   Category = c("A", "A", "B", "B", "A",
9               "B", "A", "B", "A", "B")
10 )
11 # Remove irrelevant ID
12 data_simplified <- data[, c(
13   "Age",
14   "Income",
15   "Score",
16   "Experience",
17   "Category"
18 )]
19 > print(data_simplified)

```

	Age	Income	Score	Experience	Category
1	21	25	55	1	A
2	23	30	60	2	A
3	25	35	65	3	B
4	28	42	70	4	B
5	30	48	72	5	A
6	32	52	76	6	B
7	35	60	80	7	A
8	38	68	84	8	B
9	40	75	88	9	A
10	42	80	92	10	B



Encoding six variables in a single chart for only 10 observations is generally too complex for a general audience.

The internal ID number should not be visualized because it is only an identifier and has no analytical meaning.

Including too many variables can:

- Increase cognitive load.
- Make the chart difficult to interpret.
- Distract the viewer from the important relationship.
- Make patterns harder to identify.